

Update on Standard Methods for the Examination of Water and Wastewater, 23rd Edition

and what's new for Part 4000

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Thank You

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Standard Methods

for the Examination of
Water and Wastewater™

23RD EDITION

Edited by

Rodger B. Baird

Andrew D. Eaton

Eugene W. Rice

American Public Health Association®

American Water Works Association®

Water Environment Federation®

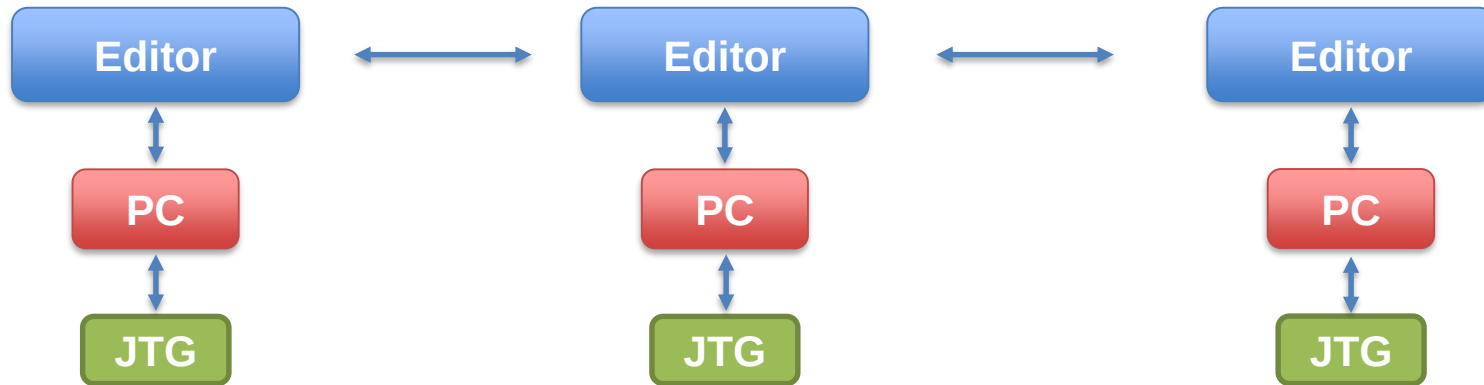
23rd Edition

Standard Methods is a joint effort of these societies

American Public Health Association (APHA)

American Water Works Association (AWWA)

Water Environment Federation (WEF)



Parts or Chapters of Standard Methods

Part 1000 Quality Assurance

Part 2000 Physical and Aggregate Properties

Part 3000 Metals

Part 4000 Inorganic Nonmetallic Constituents

Part 5000 Aggregate Organic Constituents

Part 6000 Individual Organic Compounds

Part 7000 Radioactivity

Part 8000 Toxicity

Part 9000 Microbiological Examination

Part 10000 Biological Examination

Part 1000 – Quality Assurance

Revised 1020 QA/QC for more consistency with:

- 2020
- 3020
- 4020
- 5020
- 6020



Revisions to 1040 – Method Development and Evaluation

Part 2000 – Physical and Aggregate Properties

New Method 2510C Total Intensity Odor – numerically rate perceived odor

NEW

Clarification to Method 2330 – Calcium Carbonate Saturation

Clarifications and minor additions to:

- **2540 – Solids**
 - **Quality control**
 - **Sources of Error**
 - **Interferences**



Part 3000 – Metals

Revised 3020 QA/QC



Part 4000 – Inorganic Nonmetallic Constituents

Revised 4020 QA/QC

Revised 4500-CN⁻

- Sampling, preservation, pretreatment
- Interferences

Revised 4500-NO₃⁻

- Reagent preparation, calibration, interferences

NEW

New Method 4500-O Dissolved Oxygen (optical probe)

- 46 PT samples, 2 matrices, 3 manufacturer's probes
- 85 side by side with membrane probe



ERP

Part 5000 – Aggregate Organic Constituents

Revised 5020 QA/QC

Update 5210 BOD

- Plastic bottles
- Optical probe
- pH adjust 6.5 – 7.5
- 2 or more blanks < 0.2 mg/L

Update 5310 TOC

- Remove 5310C
- Update QC criteria – moved to Introduction

Update 5910 UV- Absorbing Organic Constituents

- Buffer preparation and Sample preparation updated



Part 6000 – Individual Organic Compounds

Revised 6020 QA/QC

NEW

New Method 6810 – Pharmaceuticals and Personal Care Products

- 13 target compounds
- LCMSMS
- Single lab study – reagent water, drinking water, wastewater
- Collaborative study – 5 laboratories



Part 7000 – Radioactivity

Revised 7010 Introduction

- Reflect current regulations
- Reference sources for standards
- Discuss radioactive equilibria

Revised 7020 QC slightly

Revised 7040 Facilities

New method in online version – 7110D

NEW

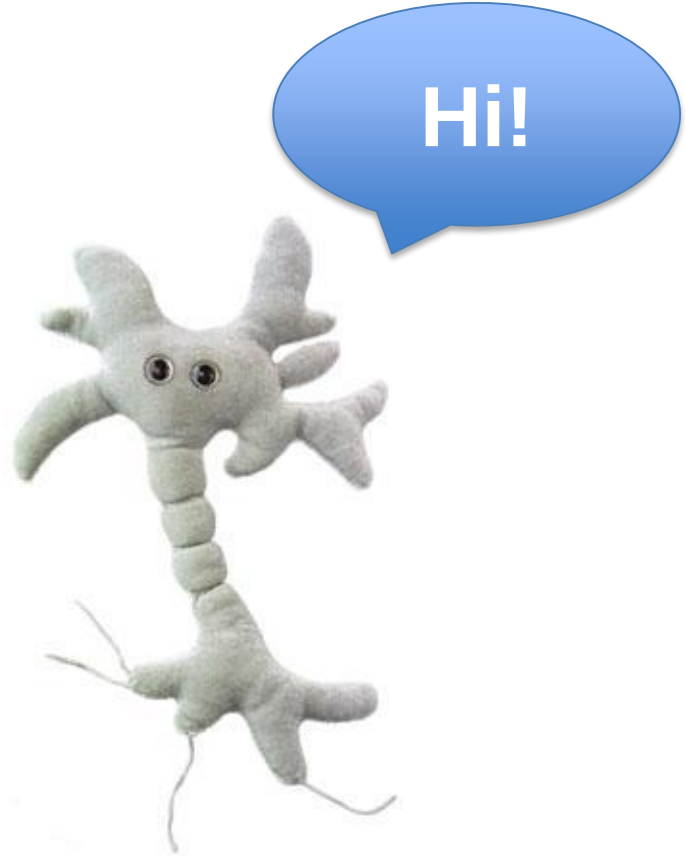
- Liquid Scintillation Method for Gross Alpha-Beta
- Works in high TDS matrices
- SDWA Expedited Method



Part 8000 – Toxicity

Revised:

- 8010 introduction growth equation
- 8020 – adds residual chlorine check
- Revisions and updates to all sections



Part 9000 – Microbiological Examination

Updated:

- 9020 QA/QC,
- 9030 Apparatus,
- 9040 Washing and Sterilization,
- 9050 Culture and Media
- 9060 Sample collection

New information on:

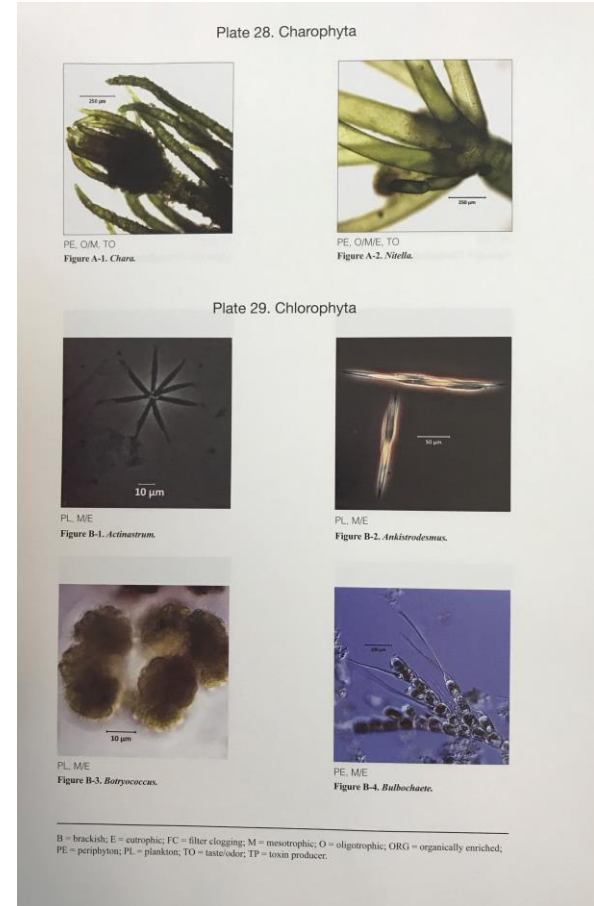
- 9215 Heterotrophic Plate Count
- 9221 Multiple Tube Fermentation
- 9222 Membrane Filtration
- 9223 Enzyme Substrate Coliform
- 9230 Fecal Enterococcus/Streptococcus
- 9250 Actinomycetes
- 9610 Fungi



Part 10000 –Biological Examination

10900 Aquatic Organisms –

Color Algae drawings replaced with
Color Algae photographs



The gold standard for over
100 years

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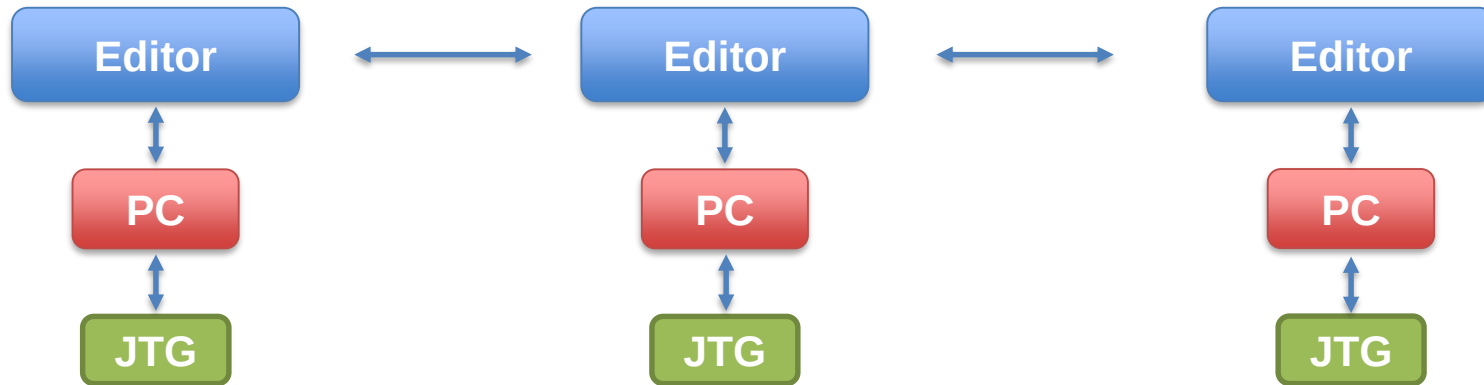
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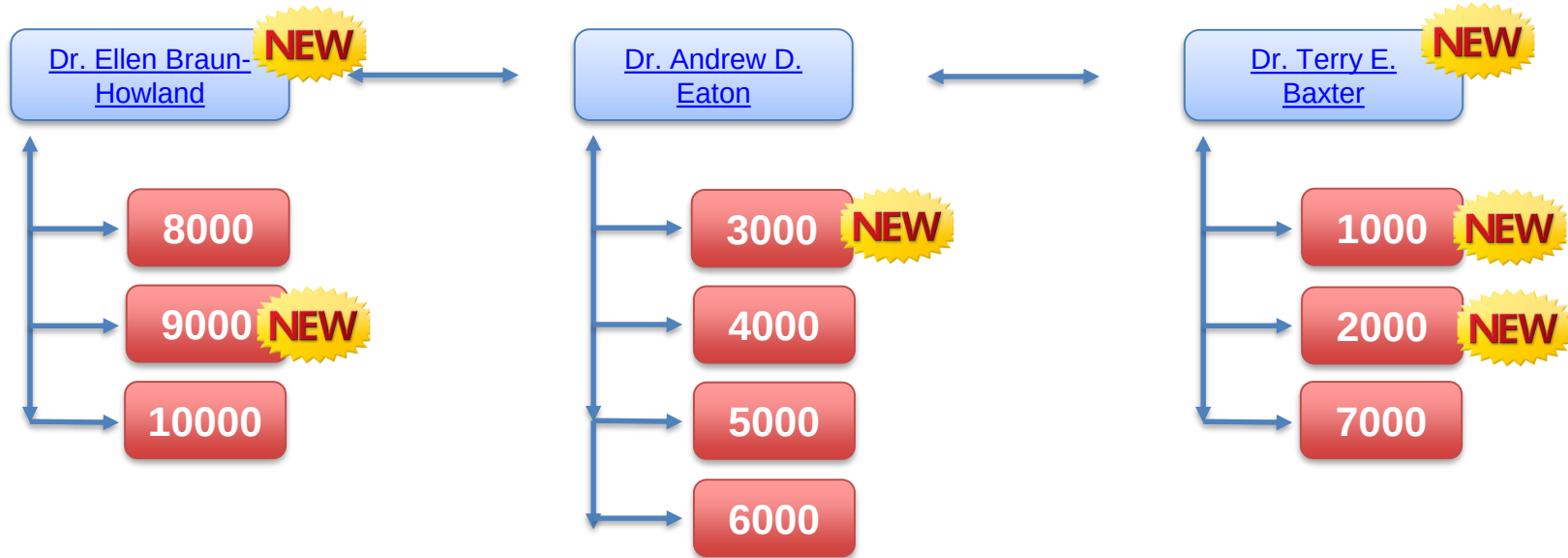


The Standard Methods Organization

American Public Health
Association (APHA)

American Water Works
Association (AWWA)

Water Environment Federation
(WEF)



Nathan Edman, E-mail: nedman@awwa.org) is Standard Methods Manager and acts as secretary to the JEB.



Part 4000 – Inorganic Nonmetallic Constituents, what's new?

Inevitable editorial changes

Modify existing methods

Develop and validate new methods



Modify existing Methods

- Technical change requires validation study
- Limited data in some
- Equivalency?



Chloride methods

- Titration methods and IC can disagree
- Interferences in titration
- Compare methods
- Collaborative study

Lab	4500-CL C	4500-CL D	4110B
1	3		
2	40	20	
3	40	61	
4	13		
5			11
6			17

Chlorine – low level amperometric

- Used in on-line analyzers
- 10 ppb ML listed in method
- Need collaborative study to validate new ML



Nitrate – UV method

- May be applicable to wastewater
- JTG to draft modified method
- Needs collaborative study for wastewater



Silica – Manual molybdenum blue

- Being used as test kits (in Drinking water)
- Different complex reagents
- Collaborative study (equivalency)
- Allow alternative complex reagents in text

Sulfite – iodometric titration

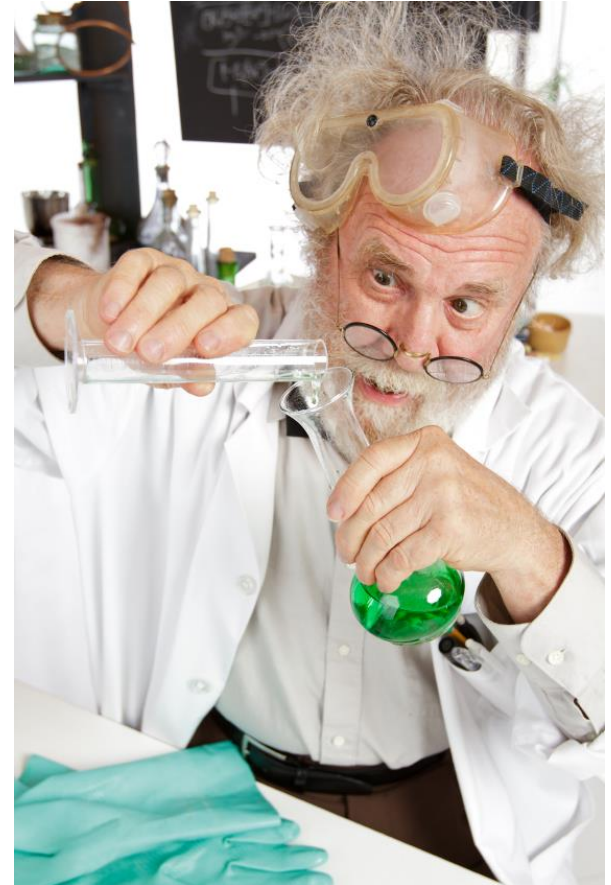
- Lot's of questions regarding Normality and the calculation
- JTG to clarify method
- Perform collaborative study

Sulfide – manual methylene blue

- Very old test with minimal data
- JTG to test new (single) reagent recipe
- If successful, re-write and collaborative study

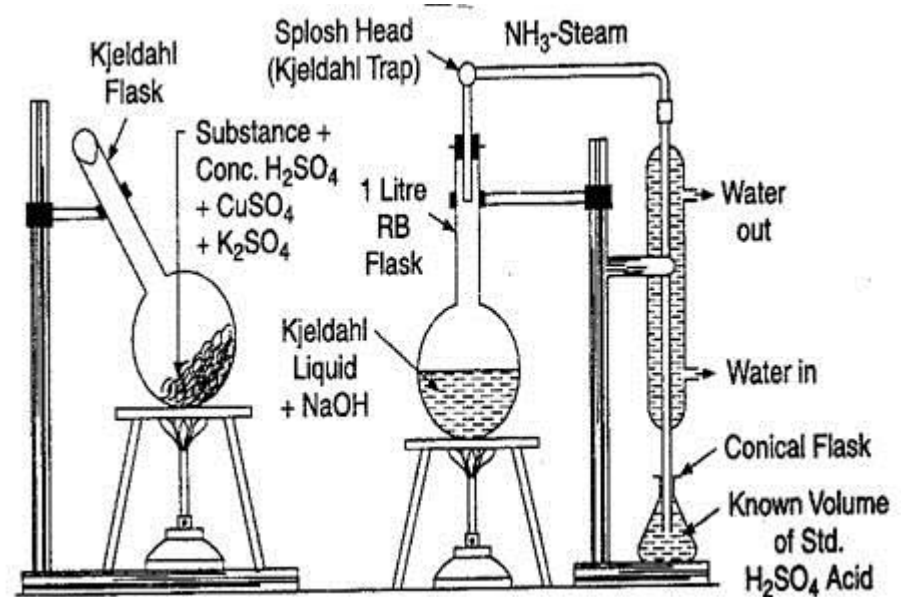
Sulfate – manual turbidimetric

- Adds BaCl_2 with a “scoop”
- Acid drainage samples (high metals) interfere
- Test interference removal
- Test 10% BaCl_2 solution with suspending agent
- Re-write method, collaborative study



Total Kjeldahl Nitrogen – with block digestion

- Already a TKN method
- No block digestion
- Change Nicotinic Acid to Glycine
- Need to prepare draft
- Need single lab study



New Methods



- Literature search
- Draft method
- Single lab data
- Ruggedness testing
- Collaborative study

Cyanide methods – gas diffusion

- EPA, ISO, and ASTM method exist
- Free, available, and total cyanide planned
- New interference studies planned (compare to WAD and CATC)
- More holding time
- Limited collaborative study (?)

Nitrate method – reductase by manual colorimetry

- ATP, and ASTM method exist for discrete analyzers
- Draft Written
- New collaborative study done
- Will ballot soon

Total Nitrogen – HTCO with chemiluminescence detection

- ASTM method
- Draft Standard Methods Written (kind of)
- Collaborative study done
- Will ballot soon



Total Nitrogen – Persulfate with nitrate by reductase

- Oxidation procedure already
- Waiting for nitrate method
- Need to show equivalency with TN
- Collaborative study planning

What is method validation anyway?



Described in SM 1040 B



- Single Operator Tests
- Analysis of real matrices
- Ruggedness Testing
- Collaborative Study

Single operator tests

- MDL
- Calibration Model
- Replicates at each concentration
- Matrix spikes, with replicates, at each concentration
- Comparison with existing method (equivalency test)

Method Ruggedness

Factor	Nominal	Variation
Mixing Time	10 minutes	12 minutes
Sample Size	5 grams	10 grams
Acid Concentration	1M	1.1 M
Heat to	100 °C	95 °C
Hold heat for	5 minutes	10 minutes
Stirring	yes	no
pH adjust	6.0	6.5

Collaborative Study – variables tested

- Laboratory – at least three
- Apparatus – model and manufacturer differences
- Concentration dependency – three or more concentrations
- Matrix effects – matrices of concern (at least three)

Example = 5 labs, 4 concentrations in triplicate



Thank You

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